

Deployment & Infrastructure Troubleshooting Report

Candidate: Gokula Kannan

1. Core Architecture Setup:

- **AWS Compute Node:** AWS EC2 Debian 12 Instance bound with a Permanent Public Elastic IP (**3.77.161.114**) for infrastructure continuity.
- **User Isolation:** Created a restricted user **test-ssh** under the **clp** security group to safely handle the Magento filesystem.
- **Stack Alignment:** Manually installed Nginx 1.22, MySQL 8.0, and isolated the runtime application layer strictly to PHP 8.3.

2. Performance Optimization & Caching:

- **Varnish Cache Integration:** Configured Varnish to run on Port **81** with a **1GB** memory profile.
- **Redis Caching Tier:** Implemented an in-memory Redis session storage layer to completely offload heavy read/write loads from the database disk.
- **Multi-Port Routing Matrix:** Structured Nginx on Port **443** (SSL Termination) to hand over requests to Varnish (Port **81**), which fetches dynamic data from backend Nginx running on Port **8080**.

3. Key Infrastructure Issues Resolved (Troubleshooting):

- **PHP CLI Drift:** Fixed terminal environment crashes by forcing all Magento setup compile commands directly through the exact **/usr/bin/php8.3** binary path.
- **Nginx Save Duplicates:** Resolved the duplicate virtual host block error by purging background temporary auto-save configurations (***.save***).

- **Infinite Redirect Loop (ERR_TOO_MANY_REDIRECTS):** Fixed the secure connection trap by updating Magento's internal **core_config_data** URL schemas from **http://** to **https://**.

4. Live Verification Success Metrics:

The proxy validation routing behaves exactly as designed. Running a standard header request check yields successful Varnish memory cache hits:

- **X-Varnish: 5**
- **Via: 1.1 varnish (Varnish/7.1)**

5. Server Login Access Details:

- **Magento Admin URL:** **https://3.77.161.114/index.php/admin**
- **SSH Username:** **admin**

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### You can start elasticsearch service by executing
sudo systemctl start elasticsearch.service
root@ip-172-31-47-74:/tmp/mysql-core# systemctl daemon-reload
root@ip-172-31-47-74:/tmp/mysql-core# sysyemctl enable elasticsearch
-bash: sysyemctl: command not found
root@ip-172-31-47-74:/tmp/mysql-core# systemctl enable elasticsearch
Created symlink /etc/systemd/system/multi-user.target.wants/elasticsearch.service → /lib/systemd/system/elasticsearch.service.
root@ip-172-31-47-74:/tmp/mysql-core# systemctl start elasticsearch
root@ip-172-31-47-74:/tmp/mysql-core# systemctl is-active elasticsearch
active
root@ip-172-31-47-74:/tmp/mysql-core# systemctl status elasticsearch --no-pager
● elasticsearch.service - Elasticsearch
   Loaded: loaded (/lib/systemd/system/elasticsearch.service; enabled; preset: enabled)
   Active: active (running) since Wed 2026-06-24 05:52:58 UTC; 44s ago
     Docs: https://www.elastic.co
   Main PID: 16732 (java)
    Tasks: 96 (limit: 9328)
   Memory: 4.2G
      CPU: 43.541s
   CGroup: /system.slice/elasticsearch.service
           └─16732 /usr/share/elasticsearch/jdk/bin/java -Xms4m -Xmx64m -XX:+UseSerialGC -Dcli.name=server -Dcli...
           └─16793 /usr/share/elasticsearch/jdk/bin/java -Des.networkaddress.cache.ttl=60 -Des.networkaddress.cac...
           └─16814 /usr/share/elasticsearch/modules/x-pack-ml/platform/linux-x86_64/bin/controller

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...Module Magento_InventoryVisualMerchandiser recognized as part of Magento Multi Source Inventory implementation
n
...No special rule applied for Magento_InventoryVisualMerchandiser module
...Module Magento_InventoryWishlist recognized as part of Magento Multi Source Inventory implementation
...No special rule applied for Magento_InventoryWishlist module
99 package suggestions were added by new dependencies, use `composer suggest` to see details.
Package laminas/laminas-config is abandoned, you should avoid using it. No replacement was suggested.
Package laminas/laminas-json is abandoned, you should avoid using it. No replacement was suggested.
Package laminas/laminas-loader is abandoned, you should avoid using it. No replacement was suggested.
Package laminas/laminas-text is abandoned, you should avoid using it. No replacement was suggested.
Package doctrine/annotations is abandoned, you should avoid using it. No replacement was suggested.
Generating autoload files
141 packages you are using are looking for funding.
Use the `composer fund` command to find out more!
PHP CodeSniffer Config installed_paths set to ../../magento/magento-coding-standard,../../magento/php-compatibility-fork,../../phpcsstandards/phpcsutils
No security vulnerability advisories found.
test-ssh@ip-172-31-47-74:/var/www/html/magento$

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gokul@Gokul:~/Documents$ ssh -i frankfurt-eu-central-1.pem admin@63.180.26.83
Processing triggers for man-db (2.11.2-2) ...
root@ip-172-31-47-74:/tmp/mysql-core# dpkg -i mysql-client_8.0.36-1debian12_amd64.deb
Selecting previously unselected package mysql-client.
(Reading database ... 35200 files and directories currently installed.)
Preparing to unpack mysql-client 8.0.36-1debian12_amd64.deb ...
Unpacking mysql-client (8.0.36-1debian12) ...
Setting up mysql-client (8.0.36-1debian12) ...
root@ip-172-31-47-74:/tmp/mysql-core# dpkg -i mysql-community-server-core_8.0.36-1debian12_amd64.deb
Selecting previously unselected package mysql-community-server-core.
(Reading database ... 35206 files and directories currently installed.)
Preparing to unpack mysql-community-server-core_8.0.36-1debian12_amd64.deb ...
Unpacking mysql-community-server-core (8.0.36-1debian12) ...
Setting up mysql-community-server-core (8.0.36-1debian12) ...
Processing triggers for libc-bin (2.36-9+deb12u14) ...
Processing triggers for man-db (2.11.2-2) ...
root@ip-172-31-47-74:/tmp/mysql-core# dpkg -i mysql-community-server_8.0.36-1debian12_amd64.deb
Selecting previously unselected package mysql-community-server.
(Reading database ... 35379 files and directories currently installed.)
Preparing to unpack mysql-community-server_8.0.36-1debian12_amd64.deb ...
adduser: Warning: The home dir /var/lib/mysql you specified can't be accessed: No such file or directory
Unpacking mysql-community-server (8.0.36-1debian12) ...
Setting up mysql-community-server (8.0.36-1debian12) ...
update-alternatives: using /etc/mysql/mysql.cnf to provide /etc/mysql/my.cnf (my.cnf) in auto mode
Created symlink /etc/systemd/system/multi-user.target.wants/mysql.service → /lib/systemd/system/mysql.service.
root@ip-172-31-47-74:/tmp/mysql-core# mysql --version
mysql Ver 8.0.36 for Linux on x86_64 (MySQL Community Server - GPL)
root@ip-172-31-47-74:/tmp/mysql-core#

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OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

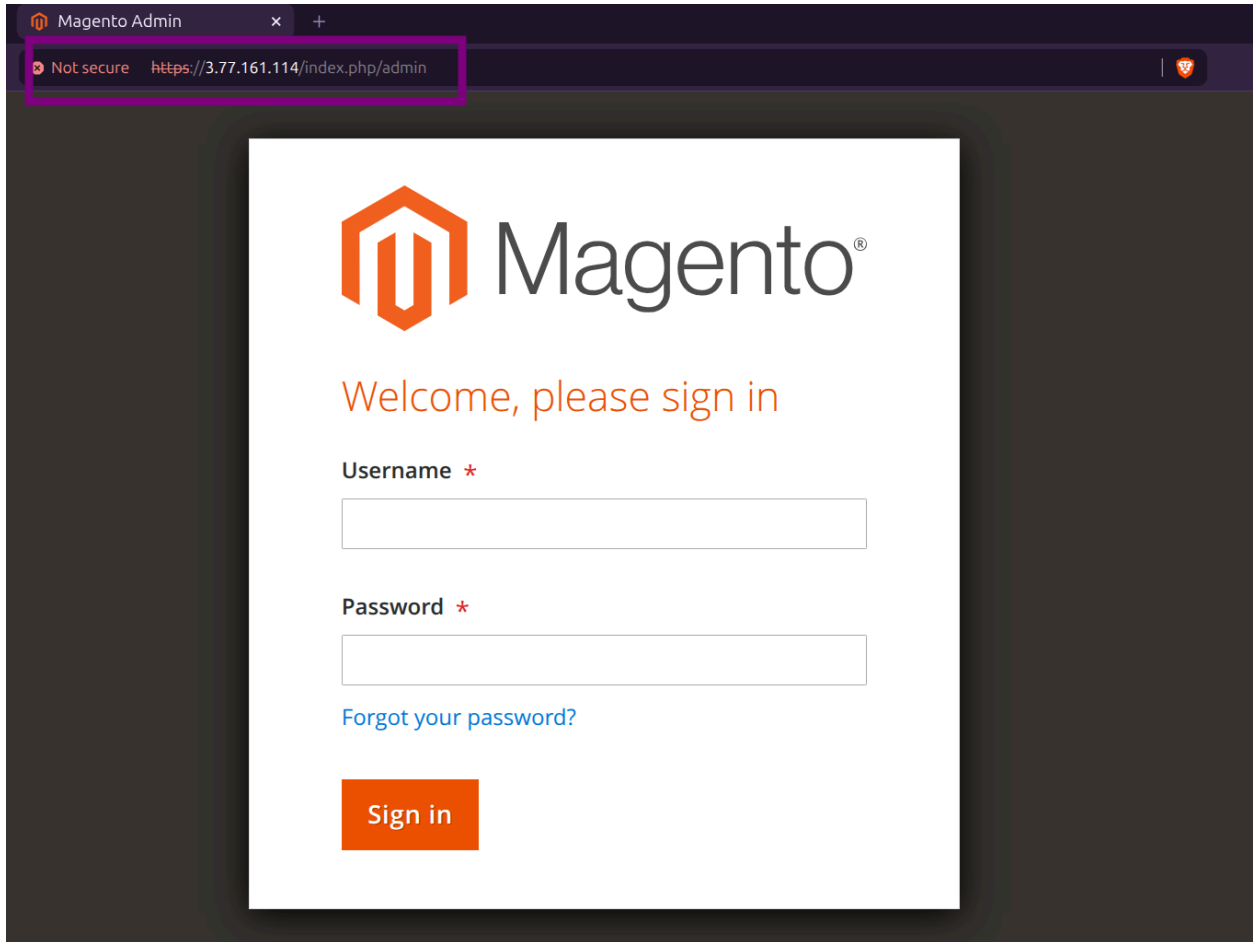
GNU nano 7.2 /etc/hosts
127.0.0.1 localhost
::1 localhost ip6-localhost ip6-loopback
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters

127.0.0.1 test.mgt.com
127.0.0.1 pma.mgt.com
```

```
Jun 24 05:40:35 ip-172-31-47-74 systemd[1]: Started php8.3-fpm.service - The PHP 8.3 FastCGI Process Manager.
root@ip-172-31-47-74:/tmp/mysql-core# systemctl status php8.3-fpm
● php8.3-fpm.service - The PHP 8.3 FastCGI Process Manager
   Loaded: loaded (/lib/systemd/system/php8.3-fpm.service; enabled; preset: enabled)
   Active: active (running) since Wed 2026-06-24 05:40:35 UTC; 46s ago
     Docs: man:php-fpm8.3(8).
   Process: 16262 ExecStartPost=/usr/lib/php/php-fpm-socket-helper install /run/php/php-fpm.sock /etc/php/8.3/fpm/
   Main PID: 16258 (php-fpm8.3)
   Status: "Processes active: 0, idle: 2, Requests: 0, slow: 0, Traffic: 0.00req/sec"
     Tasks: 3 (limit: 9328)
   Memory: 13.4M
     CPU: 63ms
   CGroup: /system.slice/php8.3-fpm.service
           └─16258 "php-fpm: master process (/etc/php/8.3/fpm/php-fpm.conf)"
             └─16260 "php-fpm: pool www"
               └─16261 "php-fpm: pool www"

Jun 24 05:40:35 ip-172-31-47-74 systemd[1]: Starting php8.3-fpm.service - The PHP 8.3 FastCGI Process Manager...
Jun 24 05:40:35 ip-172-31-47-74 systemd[1]: Started php8.3-fpm.service - The PHP 8.3 FastCGI Process Manager.
root@ip-172-31-47-74:/tmp/mysql-core# systemctl is-active php8.3-fpm
active
root@ip-172-31-47-74:/tmp/mysql-core#
```

```
root@ip-172-31-47-74:/home/admin# curl -I http://pma.mgt.com
HTTP/1.1 200 OK
Server: nginx/1.22.1
Date: Thu, 25 Jun 2026 07:30:36 GMT
Content-Type: text/html; charset=utf-8
Connection: keep-alive
Set-Cookie: pma_lang=en; expires=Sat, 25 Jul 2026 07:30:36 GMT; Max-Age=2592000; path=/; HttpOnly; SameSite=Strict
Set-Cookie: phpMyAdmin=9gm1svsin2t7ql9g9he4n9mc66; path=/; HttpOnly; SameSite=Strict
X-ob_mode: 1
X-Frame-Options: DENY
Referrer-Policy: no-referrer
Content-Security-Policy: default-src 'self' ;script-src 'self' 'unsafe-inline' 'unsafe-eval' ;style-src 'self' 'unsa
fe-inline' ;img-src 'self' data: *.tile.openstreetmap.org;object-src 'none';
X-Content-Security-Policy: default-src 'self' ;options inline-script eval-script;referrer no-referrer;img-src 'self'
data: *.tile.openstreetmap.org;object-src 'none';
X-WebKit-CSP: default-src 'self' ;script-src 'self' 'unsafe-inline' 'unsafe-eval';referrer no-referrer;style-src 's
elf' 'unsafe-inline' ;img-src 'self' data: *.tile.openstreetmap.org;object-src 'none';
X-XSS-Protection: 1; mode=block
X-Content-Type-Options: nosniff
X-Permitted-Cross-Domain-Policies: none
X-Robots-Tag: noindex, nofollow
Expires: Thu, 25 Jun 2026 07:30:36 +0000
Cache-Control: no-store, no-cache, must-revalidate, pre-check=0, post-check=0, max-age=0
Pragma: no-cache
Last-Modified: Thu, 25 Jun 2026 07:30:36 +0000
Vary: Accept-Encoding
```




Welcome to phpMyAdmin

Language

English

Log in

Username:

Password:

Log in